

Timothy-Joseph Godwin

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Education

University of Birmingham

M.Eng Computer Science with Software Engineering

Sept 2023 – present

Birmingham, UK

- **Relevant Coursework:** Data Structures & Algorithms, Object Oriented Programming, Software Engineering and Professional Practices, Team Project, Artificial Intelligence, Operating Systems & Systems Programming

Kaplan International College, London

Foundation Certificate in Science and Engineering

Jan 2023 – June 2023

London, UK

- **Relevant Coursework:** Mathematics, Information Technology, Extended Project (Research)

Experience

London Success Academy

Data Engineering Intern

Sept 2025 – Oct 2025

London, UK

- Built enterprise-scale ETL pipeline processing 300,000+ records using Python, Apache Spark, and GCP, achieving 80% storage optimisation through Parquet format with real-time API integration
- Orchestrated production data workflows using Apache Airflow on Google Cloud Composer with automated scheduling, BigQuery metadata tracking, and data validation ensuring 99%+ pipeline reliability
- Designed BigQuery data warehouse with partitioned/clustered tables processing \$15.3M transaction volume across 5 currencies, implementing query optimisation reducing costs by 80% through partition pruning
- Deployed distributed Apache Spark processing on GCP Dataproc cluster, scaling from single-machine to 3-node architecture, processing 95,000+ clickstream events and 50,000+ financial transactions with zero data loss

Uptrail

Data Analysis Intern

July 2025 – Aug 2025

London, UK

- Cleaned and analysed 2,992 sales records across 499 customers to identify top loyalty tiers and delivery optimisation opportunities (41% delay reduction potential) through 6 data visualisations and strategic analysis
- Developed Scikit-learn logistic regression model on 1,494 streaming customers achieving 76.6% accuracy, identifying mobile usage and geographic patterns as key churn drivers for targeted retention strategies
- Created interactive Tableau dashboard integrating 3 datasets (15,822 records) with 6 calculated KPIs, enabling executive strategic planning through geographic mapping and customer segmentation analysis

Projects

Campus Events Web Application

JHipster, Angular, PostgreSQL

- Developed a full-stack web application for university students to discover and review campus events and societies as part of a 7-member team using Agile methodologies
- Led CI/CD pipeline implementation using Azure for deployment, Docker containerisation, and GitHub Actions runners
- Engineered RESTful APIs with Spring Boot and resolved database migration issues using Liquibase

Airline Crew Scheduling Optimisation

Python, NumPy

- Implemented and compared three metaheuristic algorithms (Simulated Annealing, Standard and Improved Binary Genetic Algorithm) for the NP-hard Set Partitioning Problem
- Achieved 30/30 feasibility across all benchmark instances with Improved BGA, incorporating pseudo-random initialisation, stochastic ranking and heuristic improvement operators from research papers
- Identified and resolved IBGA runtime bottleneck by replacing repeated matrix scans with precomputed column/row lookups and vectorised NumPy operations, evaluated over 30 independent runs per algorithm

Technical Skills

Programming Languages: Python, SQL, Java, JavaScript, TypeScript, C, Haskell, PowerShell

Data Engineering & ML: PySpark (Apache Spark), Pandas, NumPy, Scikit-learn, Tableau, ETL Pipelines

Web Technologies: React, Angular, Spring Boot, HTML/CSS, Bootstrap, Express.js, Node.js, REST APIs

Cloud & Databases: GCP (Storage, Dataproc, Composer, BigQuery), AWS, Docker, GitHub, Azure, Apache Airflow

Additional Information

Languages: English (C1), Igbo (Native), Spanish (A1)

Achievements: 2023/24 UG Outstanding Achievement Award (SRM Funded); Silver medallist in state-level Physics and Mathematics Olympiads; Top 3 graduating student & Best Student in Further Mathematics (British Spring College)